EMPLOYEE(SSN, Name, Address, Salary, Sex, Birth_Date, Dept_No, Super_SSN)

DEPARTMENT(Dept_ID, Dept_Name, Manager_SSN, Manager_Start_Date)

DEPT_LOCATION(Dept_ID, Location)

PROJECT(Proj_No, Proj_Name, Proj_Location, Proj_Cost, Start_Date, End_Date, Department_ID)

WORKS_ON(Emp_SSN, Project_No, Hours)

1.SELECT e.*

```
FROM employee e, department d
WHERE e.dept_no = d.dept_id
AND e.salary > (
   SELECT AVG(e2.salary)
   FROM employee e2, department d2
   WHERE e2.dept_no = d2.dept_id
   AND d2.dept_name = 'finance'
);
```

2. SELECT e.name, d.dept_name

```
FROM employee e, department d, works_on w, project p, department d2
WHERE e.dept_no = d.dept_id
AND e.ssn = w.em_ssn
AND w.project_no = p.proj_no
AND p.department_id = d2.dept_id
AND d2.dept_name = 'R&D'
GROUP BY e.ssn, e.name, d.dept_name
HAVING COUNT(w.project_no) > 1;
```

3. SELECT p.proj_no, p.proj_name, d.dept_name

```
FROM project p, department d
WHERE p.department_id = d.dept_id
AND p.end_date > CURDATE();
```

4. SELECT e.*

FROM employee e, employee e2, works_on w, project p

WHERE e.ssn = e2.super_ssn

AND e.ssn = w.em_ssn

AND w.project_no = p.proj_no

AND p.end_date < CURDATE()

GROUP BY e.ssn

HAVING COUNT(DISTINCT e2.ssn) > 3;

5. SELECT dep.name

FROM dependant dep, employee e, works_on w, project p

WHERE dep.emp_ssn = e.ssn

AND e.ssn = w.em_ssn

AND w.project_no = p.proj_no

AND p.end_date < CURDATE()

GROUP BY dep.name, e.ssn

HAVING SUM(p.proj_cost) >= 100000;

6. SELECT d.dept_name, e.*

FROM employee e, department d, works_on w, project p

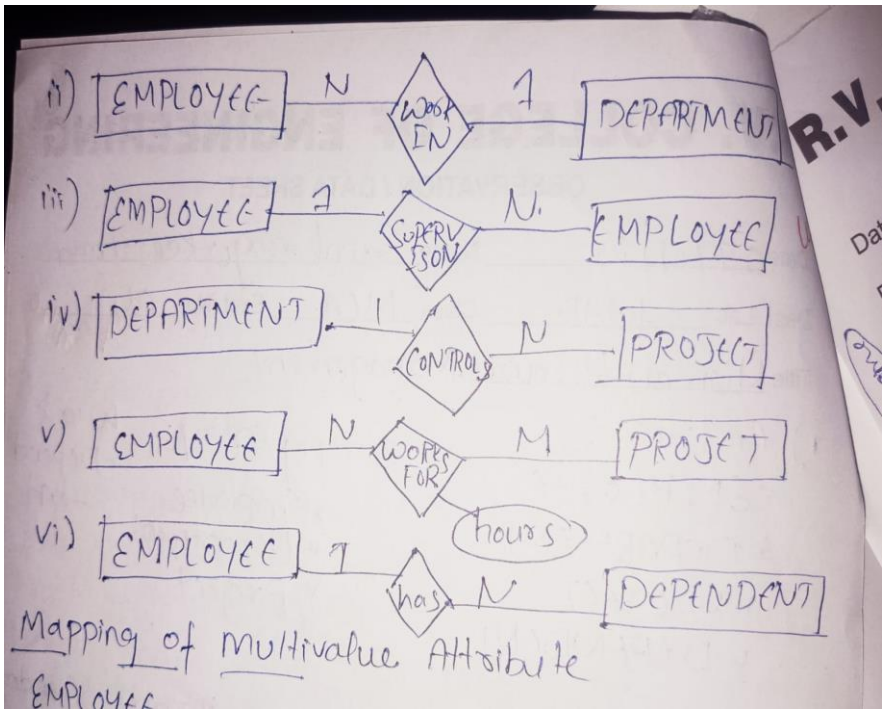
WHERE e.dept_no = d.dept_id

AND e.ssn = w.em_ssn

AND w.project_no = p.proj_no

GROUP BY e.ssn, d.dept_name

HAVING COUNT(DISTINCT p.proj_location) > 1;



Department

dept-id (P)	dname	mgr-ssn	mgr-start-date

Dept_locations

dept-no (F)	dlocation

Employee

ssn (P)	f-name	l-name	address	salary	sex	birth-date	super-ssn	dept-no (F)

Project

proj-no (P)	pname	status	budget	dept-no

Project_locations

proj-no (F)	plocation

Works_on

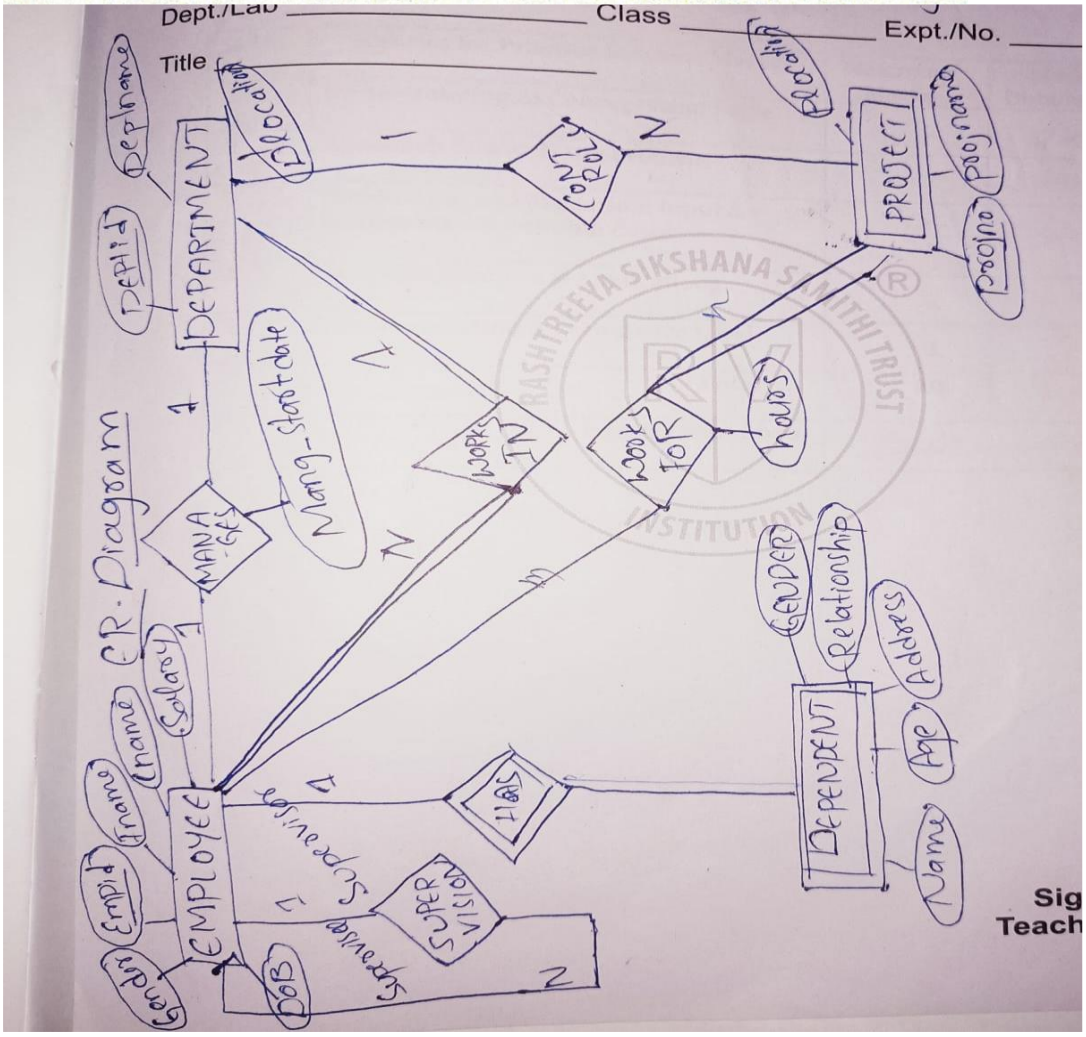
essn (F)	proj-no	hours

Dependent

essn (F)	dep-name	sex	birth-date	relationship

Dept./Lab _____ Class _____ Expt./No. _____

Title _____



Sig
Teach

a) Solar Panel: (PV_Module_Id (PK),Panel_Type,Usage_Type,Installation_Charge, Warranty_Years)
b) Distributor: (TIN (PK),Vendor_Name, Address, Contact_Details)
c) Users Place: (Place_Id,Place_Name,Address,Place_Type_
d) Relationship & Relationship Attributes
Installation: (Installation_Id, Installation_Date, Capacity, Total_Charge)

1. SELECT u.area

```
FROM user u

JOIN installation i ON i.house_no = u.h_no

JOIN solar_panel p ON i.pv_number = p.pv_no

WHERE p.pv_type = 'mono';
```

2. SELECT u.address, MAX(p.capacity)

```
FROM installation i, user u, solar_panel p

WHERE i.house_no = u.h_no

AND i.pv_number = p.pv_no

GROUP BY u.address

HAVING MAX(p.capacity) >= ALL (

   SELECT MAX(p.capacity)

   FROM installation i, user u, solar_panel p

   WHERE i.house_no = u.h_no

   AND i.pv_number = p.pv_no

   GROUP BY u.address

);
```

3. SELECT u.area

```
FROM installation i, user u, solar_panel p

WHERE i.house_no = u.h_no

AND i.install_type = p.pv_type

AND p.capacity = (SELECT MAX(capacity) FROM solar_panel);
```

4. SELECT p.pv_type, SUM(i.install_charge)

FROM installation i, user u, solar_panel p

WHERE i.house_no = u.h_no

AND i.install_type = p.pv_type

AND u.area = 'delhi'

GROUP BY p.pv_type;

5. SELECT d.tin_no, d.d_name, d.address, d.contact,

p.pv_no, p.pv_type, p.waranty, p.capacity

FROM installation i, distributor d, solar_panel p

WHERE i.tin_number = d.tin_no

AND i.pv_number = p.pv_no

AND i.install_date = (

SELECT MIN(install_date) FROM installation

);

6. SELECT p.pv_type, AVG(i.install_charge)

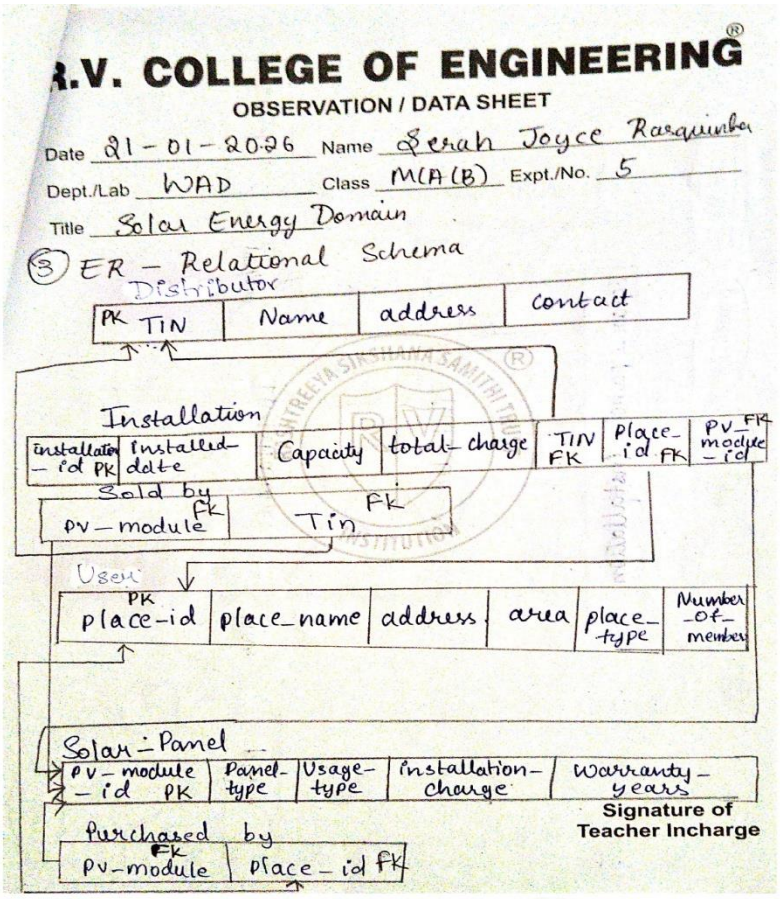
FROM installation i, user u, solar_panel p

WHERE i.house_no = u.h_no

AND i.install_type = p.pv_type

AND u.u_type = 'commercial'

GROUP BY p.pv_type;



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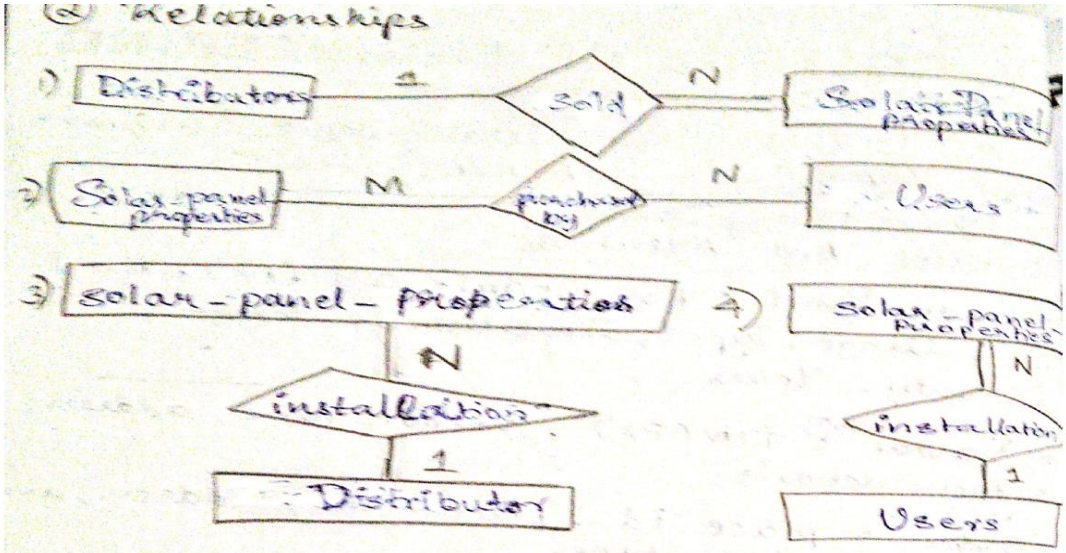


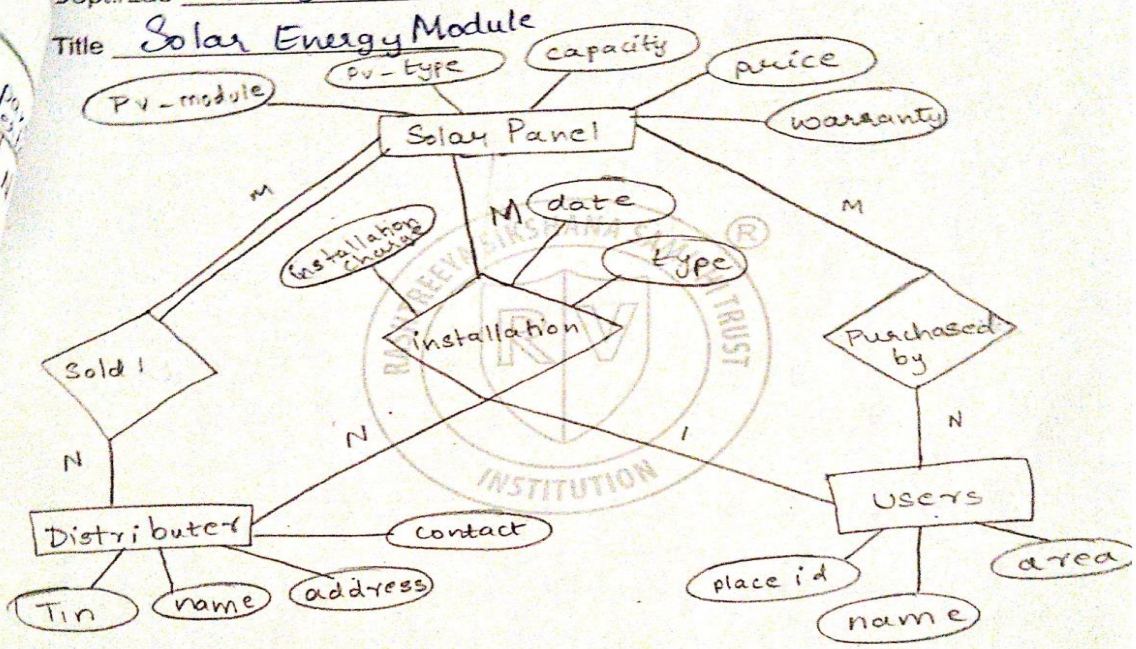
R.V. COLLEGE OF ENGINEERING®

OBSERVATION / DATA SHEET

Date 28-1-26 Name Sarah Joyce

Dept./Lab WAD Class MCA(B) Expt./No. 5

Title Solar Energy Module



Signature of Teacher Incharge

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```
let count = 0;

  document.getElementById("inventoryForm").addEventListener("submit",
function(event)   event.preventDefault();

count++;

 let table = document.getElementById("inventoryTable");

 let row = table.insertRow();

 row.innerHTML = `

        <td>${count}</td>

        <td>${name}</td>

        <td>${qty}</td>

        <td>${price}</td>

        <td><button class="btn btn-danger btn-sm"
onclick="deleteRow(this)">Delete</button></td>

      `;  document.getElementById("inventoryForm").reset();

   });

function deleteRow(button) {

      let row = button.parentElement.parentElement;

      row.remove();


let nameRegex = /^[A-Za-z ]+$/;

   let emailRegex = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

   let phoneRegex = /^[0-9]{10}$/;

   let passwordRegex = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d).{8,}$/;

 let valid = true;

 if (!nameRegex.test(name)) {

      document.getElementById("nameError").innerText = "Only letters allowed";

      valid = false;

   } else {

      document.getElementById("nameError").innerText = "";   }
```